

Marwan Abdelaal

Backend Engineer · Full-Stack Developer · AI Engineering Intern

marwan.abdelaal@outlook.com · +201111210005 · Cairo, Egypt

linkedin.com/in/marwan-abdelaal · github.com/MoscowAbdelaal · marwan-abdelaal-portfolio.vercel.app

PROFILE

3rd-year Software Engineering student at GIU (GPA: 2.4, Very Good, German Scale) with a Minor in IT Security. I build backend systems that work in the real world: production APIs, durable schedulers, idempotent pipelines, SaaS billing engines, embeddable widget platforms, visual AI workflow builders, and live-deployed AI tools. Currently interning at FlyRank AI. Led a team of 10 to ship an AI-powered MERN career platform, built Moscow AI — a Gemini-powered personal portfolio agent, and hold 20 Anthropic Academy certifications. I take ownership, ship things, and learn fast.

SKILLS

Backend:	Node.js · Express.js · REST APIs · JWT · Middleware · BullMQ · Inngest · Supabase Auth
Languages:	JavaScript · Python · Java · C# · C++ · SQL · HTML & CSS
Databases:	PostgreSQL · MongoDB · SQLite · SQL Server · Redis · Schema Design
AI & ML:	Google Gemini · Hugging Face · Ollama · OpenAI API · NLP · Prompt Engineering · Embeddings
Security:	AES-GCM · HMAC · 3DES · RSA · Diffie-Hellman · OAuth · Rate Limiting · SHA-256 PoW
Reliability:	Idempotency · Durable Scheduling · Quota Enforcement · Retry Logic · Caching · SSE
Payments:	Stripe Test Mode · Stripe CLI · Checkout · Subscription Webhooks · Token Pricing
Frontend:	React.js · Next.js · React Flow · Context API · Axios · Shadcn UI
DevOps:	Docker · Docker Compose · Git · GitHub · GitLab · CI/CD · Render
Testing:	Jest · Supertest · Postman · Playwright · Puppeteer · Automated Testing

EXPERIENCE

Backend AI Engineering Intern FlyRank AI

Jun 2026 - Present | Remote

- Shipped 3 production capstone systems: a SaaS metering and billing engine (idempotency, Stripe, quota enforcement), a multi-platform social campaign publisher (BullMQ/Redis, AES-GCM, HMAC), and an embeddable widget platform (multi-tenant, SSE, GDPR, SHA-256 PoW CAPTCHA).
- Built and deployed an AI meeting assistant on Render: Google Gemini transcript processing (10x time savings), Supabase auth, Puppeteer PDFs, Node-cron email reminders, caching, rate limiting, and Jest + Supertest tests.
- Built a visual AI workflow system (React Flow + Inngest), async background jobs API, and PDF report generator with SQL aggregation and Playwright headless rendering.
- Completed 20 Anthropic Academy certifications spanning AI fluency, Claude API, MCP, agent skills, Amazon Bedrock, and Google Cloud.

Part-Time Teaching Assistant ICTeam El Komy - IGCSE ICT

Jun 2024 - May 2026 | Cairo, Egypt

- Guided 20+ students through technical concepts across lab sessions, troubleshooting issues in real time and adapting delivery based on continuous feedback.

Event Volunteer & Usher Cairo International Film Festival & GIU Events

2024 - Present | Cairo, Egypt

- University representative at EduGate 2026. Coordinated 3+ large-scale events managing cross-functional volunteer teams under high-pressure conditions.

PROJECTS

Embeddable Widget & Lead-Capture Platform - FlyRank Capstone

Node.js · Express.js · PostgreSQL · Docker · SSE · SHA-256 PoW · GDPR | github.com/MoscowAbdelaal/flyrank-capstone-widget-platform

- Multi-tenant SaaS platform delivering embeddable widgets (signup forms, CTAs, popovers) via a single script tag — IP geo-enrichment with 4-provider fallback, honeypot spam detection, SHA-256 proof-of-work CAPTCHA, and rate limiting per IP and widget.
- Real-time dashboard via Server-Sent Events (SSE), GDPR compliance (consent, data export, right to be forgotten), production minified widget bundle, page targeting rules, and safe fire-and-forget side effects so email/webhook failures never break

submissions.

AI Meeting Assistant - Full-Stack AI-Powered Meeting Tool

React.js · Node.js · Google Gemini API · Supabase · Puppeteer · Docker · Jest · Render | github.com/MoscowAbdelaal/ai-meeting-assistant

- Full-stack AI tool processing meeting transcripts with Google Gemini to extract summaries, decisions, and action items — reducing processing time from 30 minutes to 3 minutes (10x improvement).
- Supabase auth, in-memory AI caching (24-hour TTL), rate limiting, Puppeteer PDF reports, daily Node-cron email reminders, Jest + Supertest test suite, deployed on Render via Docker.

AI Decision Flow - Visual AI Workflow Builder

React.js · Next.js · React Flow · Inngest · OpenAI API · Shadcn UI | github.com/MoscowAbdelaal/ai-decision-flow

- Drag-and-drop React Flow editor where each node maps to an LLM decision step (YES/NO), with Inngest executing each node as a durable step and dynamically traversing the graph based on AI responses.

Usage Metering & Billing Engine - FlyRank Capstone

Node.js · PostgreSQL · Docker · Stripe · Redis · Idempotency | github.com/MoscowAbdelaal/flyrank-capstone-metering-billing

- Exactly-once metering prevents double-charging, quota enforcement returns 429/402 at plan boundaries, AI token pricing handles cached/reasoning tokens, and Stripe HMAC webhooks deduplicate subscription events.

Multi-Platform Social Campaign Publisher - FlyRank Capstone

Node.js · BullMQ · Redis · PostgreSQL · Docker · Sharp · AES-GCM · HMAC | github.com/MoscowAbdelaal/flyrank-capstone-social-studio

- Blog post to full social campaign: platform-correct image variants, prompt-fragment captions, crash-safe BullMQ/Redis scheduler, HMAC-verified webhooks, and AES-GCM encrypted OAuth tokens.

GIU Nexus - AI-Powered Career & Talent Platform

React · Node.js · MongoDB · Hugging Face · JWT · GitLab | github.com/MoscowAbdelaal/GIU-Nexus

- Led 10 developers to ship a MERN platform with Hugging Face NLP embeddings, zero-shot job classification, cosine similarity recommendations, and role-based dashboards for job seekers, recruiters, and admins.

Moscow AI - Personal Portfolio AI Agent

Next.js · Google Gemini API · Prompt Engineering · Vercel | marwan-abdelaal-portfolio.vercel.app

- Gemini-powered AI agent embedded in portfolio site answering recruiter questions using verified portfolio data only, constrained via prompt engineering to prevent hallucination.

Cryptography Suite - 3DES & Diffie-Hellman/RSA

Python · Jupyter Notebook

- Triple DES from scratch (EDE, 16-round blocks) and a Diffie-Hellman/RSA secure channel proving end-to-end encryption and authentication without library shortcuts.

Clash of Clans: Heroes - 2-Player Strategy Game

Java · JavaFX · OOP · Eclipse IDE | github.com/MoscowAbdelaal/Clash-of-clans-Legends

- Complete 2-player turn-based strategy game across 3 milestones with full OOP architecture, custom exception handling, and JavaFX frontend with real-time state updates and victory detection.

CERTIFICATIONS - ANTHROPIC ACADEMY (2026)

- | | |
|--|---|
| - AI Fluency: Framework & Foundations | - Claude 101 |
| - Claude Code 101 | - Building with the Claude API |
| - Introduction to Model Context Protocol | - Model Context Protocol: Advanced Topics |
| - Introduction to Agent Skills | - Introduction to Subagents |
| - Claude with Amazon Bedrock | - Claude on Google Cloud |

20 certifications total - full list at [linkedin.com/in/marwan-abdelaal](https://www.linkedin.com/in/marwan-abdelaal)

EDUCATION

**Bachelor of Software Engineering - Informatics & Computer Science
German International University (GIU)**

Sep 2024 - Present | Cairo, Egypt

GPA: 2.4 / Very Good (German Scale) · Minor: IT Security · Expected Graduation: 2028 · Coursework: DSA, OOP, Database Systems, Data Science, Information Security, Distributed Systems, Media Informatics

LANGUAGES

- | | |
|--------------------------|--------------------------|
| - Arabic Native | - English Fluent |
| - German Beginner | - French Beginner |